EARTH LAB / SCIENTIFIC PYTHON

SpectralBridge

From flightlines to analysis-ready data

THREE INPUT STREAMS / THREE WORKFLOWS

1 NEON flightlines

Airborne hyperspectral HDF5

Download > correct > sensor bands

Flight-by-flight processing

2 Local drone imagery

Supported TIFF / HDF5

Discover > process > export

Separate local-image workflow

3 Completed flightline products

Existing rasters and tables

Stream > compare > summarize

No new download or correction

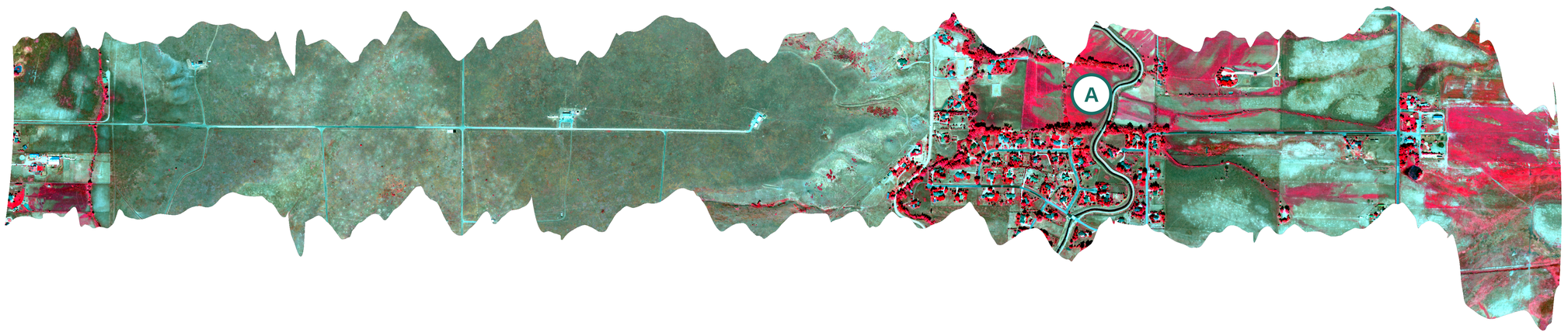
Workflow outputs: traceable rasters, tables, QA and comparison reports

REAL FLIGHTLINE / VISIBLE LIGHT

A flightline is the strip recorded along one flight path.

SAME GROUND / INFRARED SHOWN IN RED

A marks the same patch in both images.

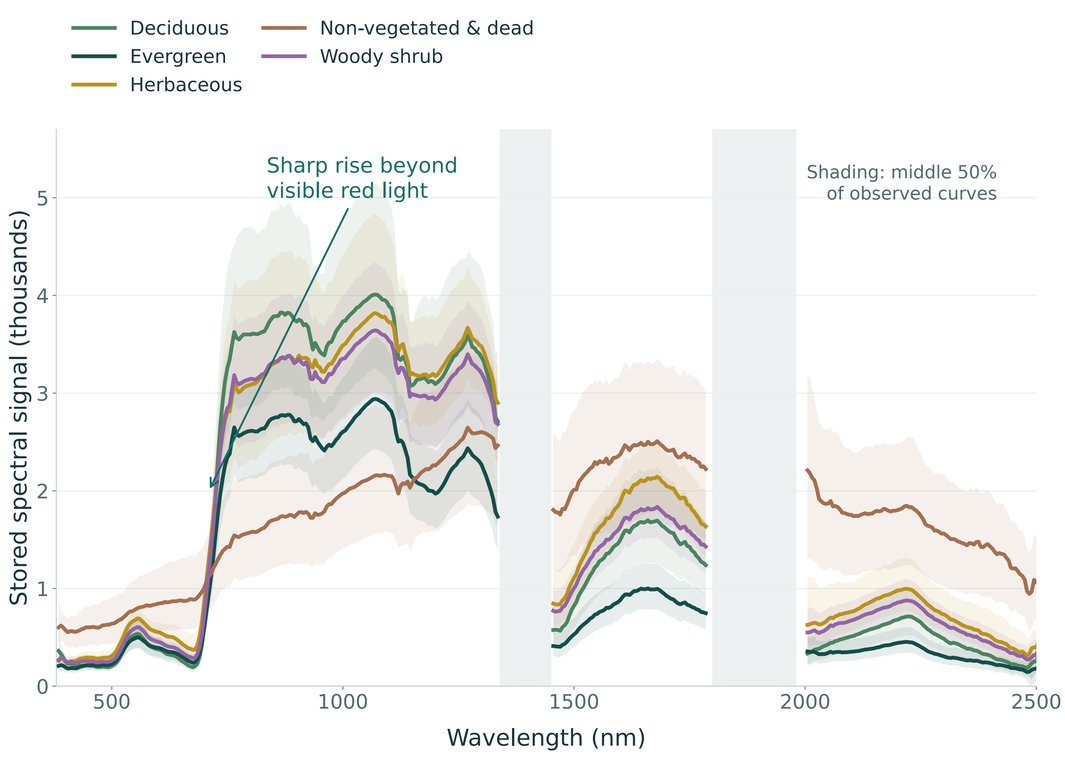


The same ground, shown with different light channels. A marks the same patch.

Red is a display color, not heat or fire. These views do not show correction improvement.

Read the spectral patterns

Groups reflect different patterns across wavelengths.

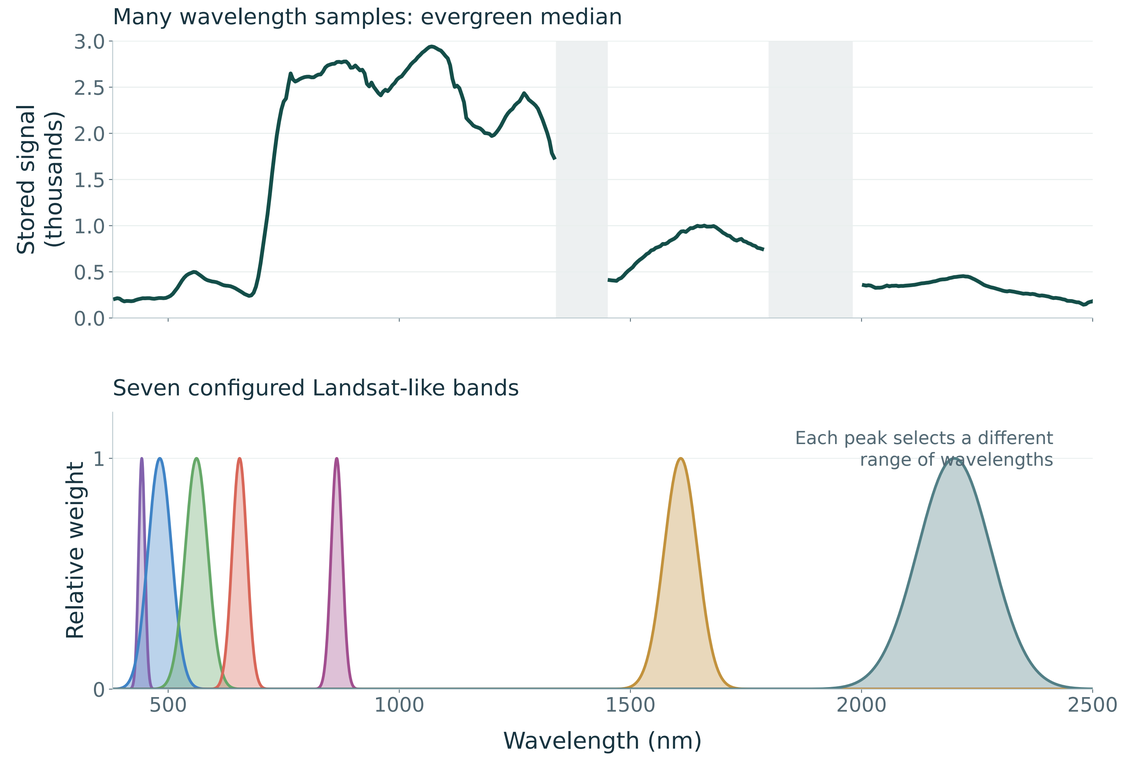


Bold curves: category medians. Bands: middle 50%.

Gray gaps omit unsupported spectral regions.

Turn spectra into sensor bands

The package weights wavelength ranges for a target sensor.



A detailed spectrum becomes fewer sensor channels.

Gaussian weights shown; not measured response curves.

01 Prepare

NEON or supported drone inputs

02 Correct

Terrain + viewing geometry

03 Translate

Configured sensor-like bands

04 Explore

Tables, QA, bulk + libraries

Restart-safe files. Chunked processing. Traceable outputs.

earthlab.github.io/spectralbridge

github.com/earthlab/spectralbridge

Spectral graphs: saved polygon/year library summaries; separate from the illustrated R10C flightline. Physical-unit scaling unresolved.

Flightline: NEON R10C, 15 September 2021, native 1 m pixels. Display: 2-98% stretch, gamma 0.8; blank exterior has no data.

Configured sensor-like bands do not establish independent sensor agreement or field cover accuracy. | Earth Lab | 27 October 2026